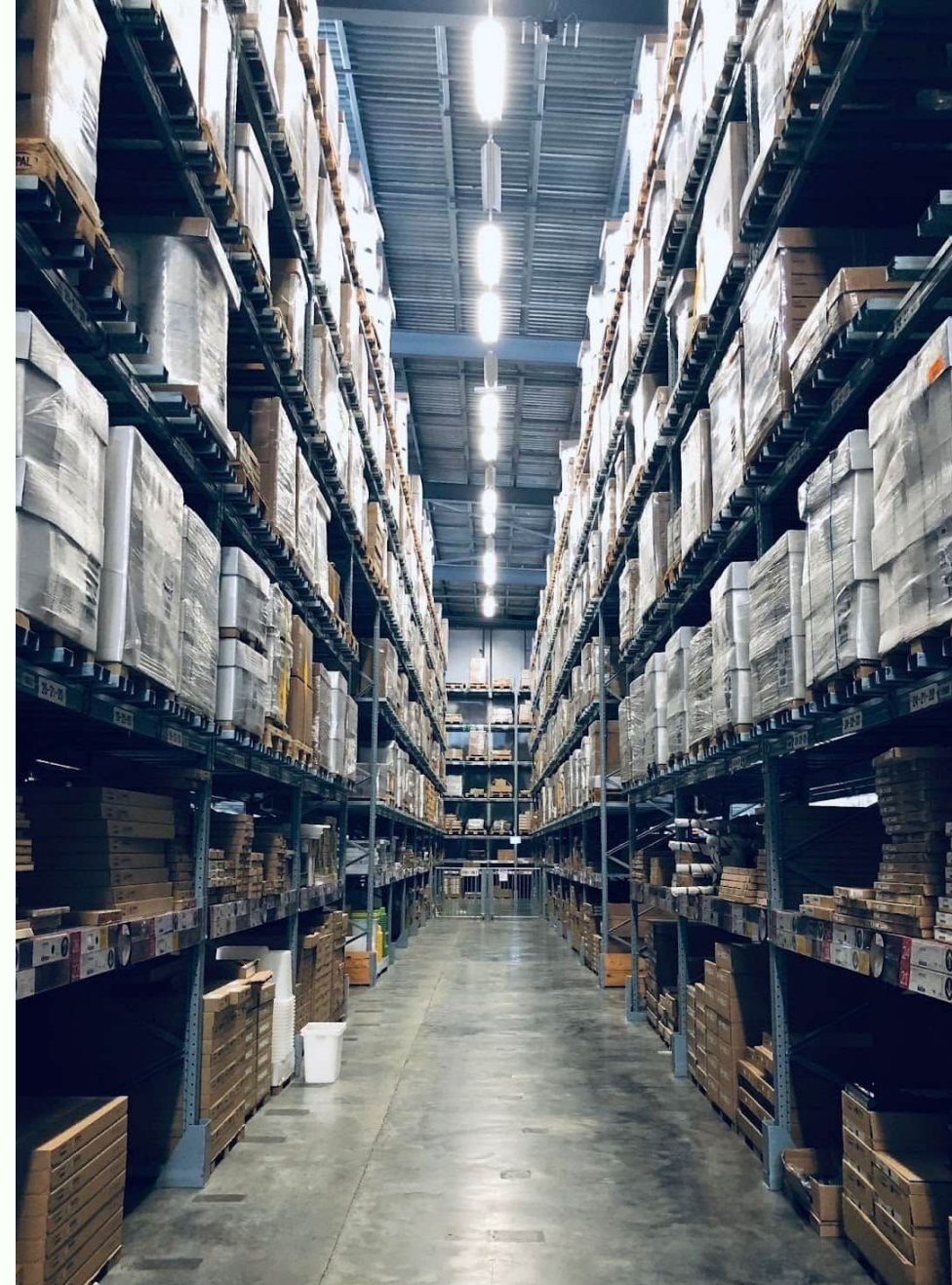


ARMY PUBLIC SCHOOL, AHMEDABAD

Predictive AI & IoT Framework for Early Post-Harvest Spoilage Detection

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This project was developed as part of our school science research program, exploring how simple technology can solve big real-world problems in Indian agriculture.



Objective

The Main Goal

Build a **"smart alarm"** for agricultural godowns that stops food waste before it happens.

Catch It Early

Detect rotting food **before it looks rotten** – using invisible signals, not human eyes.

Save Money

Give warehouse managers an **early warning** so they know exactly when their stock is about to spoil.



Area Map



Where This System Will Be Used

Our system is designed for the **large godowns and wholesale mandi warehouses** across Ahmedabad – places where tonnes of produce are stored every day.

- Large bulk storage godowns in Ahmedabad
- Wholesale mandi facilities (agricultural markets)
- Indoor warehouse complexes storing grain and produce

❏ These locations handle massive quantities of food – even small spoilage losses add up to huge waste.

Methodology



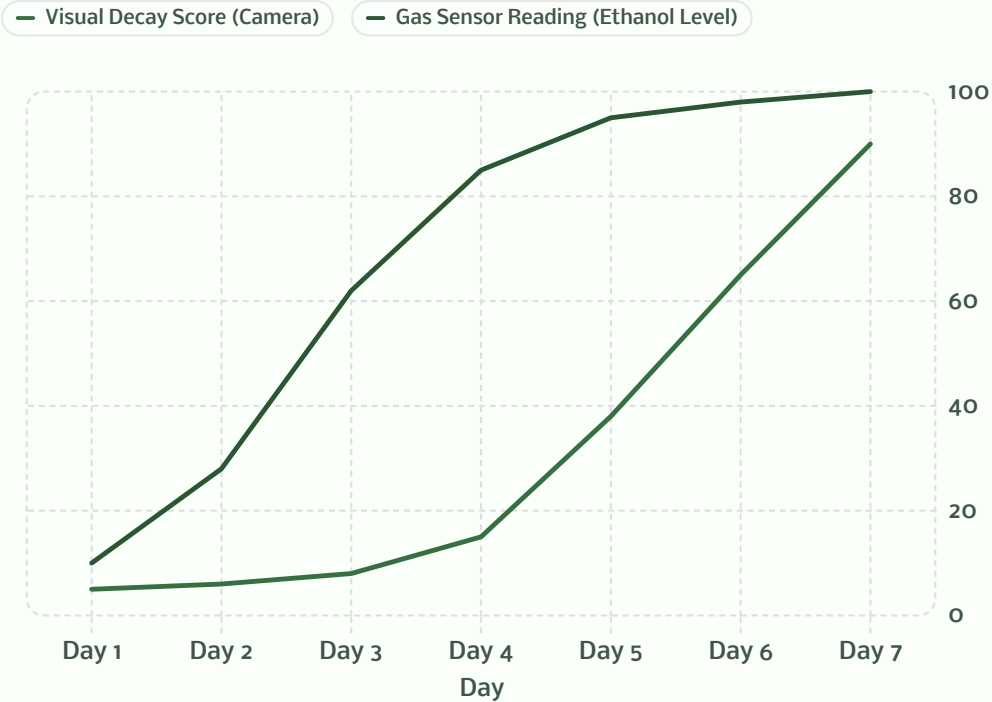
The Core Idea

The Setup

The Test

When produce starts to go bad, it **"burps" out invisible gases** like ethanol – long before it turns brown or mushy. We built a simple experiment to prove which sensor catches this signal first.

Results



What We Discovered

The electronic **"nose" is much faster** than the "eyes."

The Arduino gas sensor picked up invisible ethanol gas **well over a full day** before the AI camera could detect any visual change. The gas rises first – the colour change follows later.

⚡ 28–36 Hours

Gas sensor detected spoilage **before** the camera saw any brown spots.

Conclusion

Looking is not enough

In a massive godown, you **cannot rely on eyes alone** to know if food is still fresh. By the time you see brown spots, the rot has already spread.

Invisible gas tells the truth

Monitoring **invisible gases** is the most reliable way to predict the future health of a stored crop – it gives us a peek into the future.

The core idea is proven

Our 7-day experiment confirms it: **gas detection beats visual detection** every single time.



Solution



The Real-World Fix

Install a **low-cost sensor box** inside local godowns. When gas levels rise, the manager gets a **phone alert** – giving them a **1.5-day head start**.

✓ Result: **Massive food waste is completely prevented** – saving money, food, and farmers' hard work.

→ Alert arrives before any visible damage

→ Manager sells or moves the at-risk batch immediately

→ Rot is stopped before it spreads to the rest of the warehouse

Project Model

